

PERIPHERAL VASCULAR

Maddock, W. G. and Collier, F.A.: Peripheral vasoconstriction by tobacco and its relation to thrombo-angiitis obliterans. Ann. Surg. 98:70-81, 1933.

This study made on the effect of tobacco smoking, largely in the form of cigarettes on young smokers, demonstrated a consistent increase in blood-pressure and pulse rate and a decrease in the skin temperature of the fingers and toes. Control experiments gave definite evidence that these effects were due to active products absorbed from the tobacco smoke. Nicotine administered intravenously in quantities not greater than that theoretically absorbed in the smoking of one or two cigarettes produced comparatively analogous changes. Greater effects were noted when the subject "inhaled" while smoking rather than merely "puffing," and also with rapid smoking more than with slow smoking. The decrease in the peripheral skin temperature on smoking must be due to increased vasoconstriction.

It was pointed out the value of measurements of the skin temperature of the fingers and toes under well-controlled conditions as indications of stimulation or depression of sympathetic nervous system. In this study the decrease in the peripheral skin temperature was shown to be carried out through that system. While the peripheral vasoconstriction on smoking was usually measured only in the fingers and toes, the points of maximal response to changes in peripheral vasomotor tonus, it is undoubtedly true that vasoconstriction of skin vessels occurred to a lesser degree over the entire body. By increasing peripheral vasoconstriction smoking reduced the blood supply of the fingers and toes of the young adults studied. In some subjects the reduction lasted more than thirty minutes from the time of cessation of smoking and generally was of longer duration in the toes than in the fingers. In two cases of thrombo-angiitis obliterans, smoking produced the same cardiovascular response as in the normal subjects. The data presented by this investigation is not offered as evidence that tobacco smoking is the etiological factor in thrombo-angiitis obliterans. The occurrence of the disease in individuals who have never smoked precludes that opinion. It is interesting however, to recall that other vasoconstricting substances, pituitrin and particularly ergot, have been responsible for peripheral vascular occlusions and gangrene. The

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